Section 8.3: Feynman Rules for QCD

A Gentler Approach

Reading Session Notes — Griffiths Chapter 8

The Big Picture: QCD ≈ QED + Color

What is Section 8.3 about?

Griffiths is answering one question: *If I want to compute a QCD process the same way I compute QED processes, what do I change?*
The answer is surprisingly little. You keep the same Dirac spinors, the same gamma matrices, the same propagator structure. You make three modifications:

1. Replace the electric charge g_e with the strong coupling g_s.
2. Attach a **color label** to every quark and gluon line.
3. Add new vertices that don't exist in QED: gluon self-interactions.

That's it. Everything else is bookkeeping.

1 Group Theory Reminder: Why SU(3)?

Quick Refresher: What is SU(N)?

SU(N) is the group of $N \times N$ unitary matrices with determinant 1.

SU(2) is the symmetry group of spin. It has:

- $2^2 - 1 = 3$ generators: the Pauli matrices $\sigma^1, \sigma^2, \sigma^3$
- States live in a **2-component** space (spin up, spin down)
- The generators satisfy $[\sigma^i, \sigma^j] = 2i\epsilon^{ijk}\sigma^k$

SU(3) is the symmetry group of color. It has:

- $3^2 - 1 = 8$ generators: the Gell-Mann matrices $\lambda^1, \ldots, \lambda^8$
- States live in a **3-component** space (red, green, blue)
- The generators satisfy $[\lambda^a, \lambda^b] = 2if^{abc}\lambda^c$

Key analogy: Just as the Pauli matrices rotate spin states into each other, the Gell-Mann matrices rotate color states into each other. A gluon emission is nothing but a color rotation at a spacetime point.

The Pauli → Gell-Mann analogy

You already know the Pauli matrices well:

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

σ^1 and σ^2 are *off-diagonal* — they flip spin up ↔ spin down. σ^3 is *diagonal* — it measures which spin state you're in without changing it.

The Gell-Mann matrices are the exact same idea, extended to three states. For example:

$$\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \longleftrightarrow \quad \text{mixes red} \leftrightarrow \text{green (leaves blue alone)}$$

$$\lambda^4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad \longleftrightarrow \quad \text{mixes red} \leftrightarrow \text{blue (leaves green alone)}$$

$$\lambda^6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad \longleftrightarrow \quad \text{mixes green} \leftrightarrow \text{blue (leaves red alone)}$$

Physical meaning of the λ matrices

Each λ^a tells you **which color transition the gluon mediates**. An off-diagonal λ physically changes a quark's color (e.g., red → green). A diagonal λ leaves the color unchanged but contributes a phase.
There are 3 off-diagonal pairs $\{(\lambda^1, \lambda^2), (\lambda^4, \lambda^5), (\lambda^6, \lambda^7)\}$ — one for each of the three possible

color swaps — plus 2 diagonal matrices $\{\lambda^3, \lambda^8\}$. That gives $6 + 2 = 8$ generators total.

Structure constants f^{abc}

The commutator of two generators defines the **structure constants**:

$$[\lambda^a, \lambda^b] = 2if^{abc}\lambda^c$$

These are just numbers (fully antisymmetric in a, b, c). They encode the "algebra" of the group — how two successive color rotations combine. In QED, the analogous commutator is trivially zero because the gauge group U(1) has only one generator. That single fact — $f^{abc} \neq 0$ for SU(3) — is why QCD has gluon self-interactions and QED does not.

The nonzero structure constants are (Griffiths Eq. 8.36):

$$f^{123} = 1, \qquad f^{147} = f^{246} = f^{257} = f^{345} = f^{516} = f^{637} = \tfrac{1}{2}, \qquad f^{458} = f^{678} = \tfrac{\sqrt{3}}{2}$$

All others are zero or obtained by antisymmetry ($f^{abc} = -f^{bac}$, etc.).

2 Color States: Quarks and Gluons

Quark colors

A quark's color state is a 3-component column vector:

$$r = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad g = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad b = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

How color attaches to spinors

In QED, a quark line carries a Dirac spinor $u^{(s)}(p)$. In QCD, it carries $u^{(s)}(p) \cdot c_i$, where c_i is the color vector above.

The full quark state is a **direct product**: spacetime structure (spinor) $\otimes$ internal structure (color). These two parts are independent — the gamma matrices act on the spinor, the λ matrices act on the color vector, and they never mix.

Gluon colors

Gluons live in the **adjoint representation** of SU(3), which is 8-dimensional. Each gluon state is labeled by an index $a = 1, \ldots, 8$ corresponding to the eight generators λ^a. The gluon's color state is an 8-component column vector $\mathbf{a}$:

$$\text{Gluon in state } |1\rangle: \quad \mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \qquad \text{Gluon in state } |5\rangle: \quad \mathbf{a} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \text{etc.}$$

Why 8 gluons

Naïvely: 3 colors $\times$ 3 anticolors = 9 combinations. But one combination,

$$|9\rangle = \frac{1}{\sqrt{3}}\big(r\bar{r} + g\bar{g} + b\bar{b}\big),$$

is a **color singlet** — it's invariant under all SU(3) transformations. A singlet gluon would mediate a long-range, strong-coupling force between any two hadrons (which are themselves singlets). We don't observe this, so it doesn't exist. The remaining 8 combinations form the color octet, and those are the physical gluons.

Group theory statement: $3 \otimes \bar{3} = 8 \oplus 1$. QCD uses SU(3), not U(3), so only the octet appears.

3 The Feynman Rules: QED $\to$ QCD

The translation table

Here is the complete dictionary for going from QED to QCD at tree level:

Element	QED	QCD
Coupling constant	$g_e = \sqrt{4\pi\alpha}$	$g_s = \sqrt{4\pi\alpha_s}$
Quark spinor (incoming)	$u^{(s)}(p)$	$u^{(s)}(p)\, c_i$
Quark spinor (outgoing)	$\bar{u}^{(s)}(p)$	$\bar{u}^{(s)}(p)\, c_j^\dagger$
Mediator polarization	ε^μ	$\varepsilon^\mu\, a^\alpha$
Quark–mediator vertex	$-ig_e\gamma^\mu$	$-\dfrac{ig_s}{2}\,\lambda^a_{ji}\,\gamma^\mu$
Mediator propagator	$\dfrac{-ig_{\mu\nu}}{q^2}$	$\dfrac{-ig_{\mu\nu}\,\delta^{ab}}{q^2}$
Fermion propagator	$\dfrac{i(\not{q} + mc)}{q^2 - m^2c^2}$	same
Self-interaction vertices	none	3-gluon, 4-gluon

The fermion propagator is **identical** — gluons don't change it.

Let's unpack each new element.

3.1 Rule 1: External Lines

Every external quark line now carries a color vector alongside its spinor:

$$\begin{aligned} \text{Incoming quark:} &\quad u^{(s)}(p)\, c \\ \text{Outgoing quark:} &\quad \bar{u}^{(s)}(p)\, c^\dagger \\ \text{Incoming antiquark:} &\quad \bar{v}^{(s)}(p)\, c^\dagger \\ \text{Outgoing antiquark:} &\quad v^{(s)}(p)\, c \end{aligned}$$

Every external gluon carries a polarization vector ε^μ *and* a color vector a^α ($\alpha = 1, \ldots, 8$).

Antiquark color convention

Notice the transposition: an outgoing quark has $c^\dagger$ (a row vector), and an incoming antiquark also has $c^\dagger$. This follows the same logic as $\bar{u}$ vs. v — you work *backward* along the fermion arrow.

3.2 Rule 2: Propagators

Quark propagator: Identical to QED.

$$\frac{i(\not{q} + mc)}{q^2 - m^2c^2}$$

No color matrix appears here — the quark color is carried straight through the propagator without change.

Gluon propagator:

$$\frac{-ig_{\mu\nu}\,\delta^{ab}}{q^2}$$

The δ^{ab} is the new ingredient. It says: *the gluon emitted at one vertex must be the same type as the gluon absorbed at the other vertex.* If vertex 1 emits gluon type $a = 3$ and vertex 2 absorbs gluon type $b = 5$, then $\delta^{35} = 0$ and the diagram doesn't contribute.

Why δ^{ab}?

This is just color conservation. In QED, there's only one photon, so there's nothing to keep track of. In QCD, there are 8 gluon types, and the propagator has to ensure the same gluon flows through the internal line. The δ^{ab} does this job.

3.3 Rule 3: The Quark–Gluon Vertex

This is the heart of QCD:

QCD vertex factor

$$\text{Quark–gluon vertex:} \qquad -\frac{ig_s}{2}\,\lambda^a_{ji}\,\gamma^\mu$$

where:

- $g_s = \sqrt{4\pi\alpha_s}$ is the strong coupling constant
- λ^a_{ji} is the (j, i) element of the a-th Gell-Mann matrix
- i = color of the *incoming* quark, j = color of the *outgoing* quark
- a = color label of the gluon
- μ = Lorentz index (contracted with the gluon propagator or polarization)

How to read λ^a_{ji}

This is just a number. For example: a red quark ($i = 1$) emits a gluon of type $a = 1$ and becomes a green quark ($j = 2$). What's the color factor?
Look up the (2,1) element of λ^1:

$$\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \Longrightarrow \quad \lambda^1_{21} = 1$$

So this process is allowed (nonzero). If the quark tried to stay red ($j = 1$), we'd need $\lambda^1_{11} = 0$ — forbidden for this gluon type.

In words: The λ matrix tells you whether a given color transition is allowed for a given gluon type, and with what amplitude.

Factorization: color $\times$ spin

A crucial simplification: the vertex factor $-\frac{ig_s}{2}\lambda^a_{ji}\gamma^\mu$ splits into two independent pieces:

$$\underbrace{\lambda^a_{ji}}_{\text{color factor (just a number)}} \times \underbrace{\left(-\frac{ig_s}{2}\gamma^\mu\right)}_{\text{Dirac structure (same as QED)}}$$

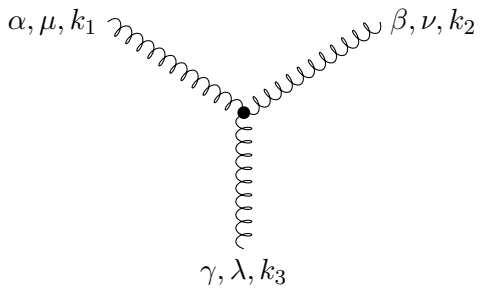
The color part and the Dirac part *never talk to each other*. You can evaluate all the spinor/trace algebra exactly as in QED, then multiply by the color factor at the end. This is enormously simplifying.

3.4 Rule 3 (continued): Gluon Self-Interaction Vertices

What's new in QCD

These vertices have **no analog in QED**. They exist because gluons carry color charge, so they can interact with each other directly. The photon is electrically neutral, so there is no photon–photon–photon vertex in QED.

Three-gluon vertex



The vertex factor is:

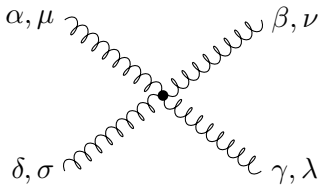
$$-g_s\, f^{\alpha\beta\gamma}\Big[g^{\mu\nu}(k_1-k_2)^\lambda + g^{\nu\lambda}(k_2-k_3)^\mu + g^{\lambda\mu}(k_3-k_1)^\nu\Big]$$

Reading the three-gluon vertex

This looks intimidating, but it has a clear structure:

- $f^{\alpha\beta\gamma}$ is the structure constant — it determines *which combinations of three gluon types are allowed*. If $f^{\alpha\beta\gamma}=0$, those three gluons don't couple.
- The bracket contains three terms, one for each pair of gluons, each involving the *difference* of two momenta. This structure is completely fixed by the gauge symmetry — you have no freedom here.
- All momenta k_1, k_2, k_3 point **into** the vertex. If one points outward in your diagram, flip its sign.

Four-gluon vertex



$$-ig_s^2\Big[f^{\alpha\beta\eta}f^{\gamma\delta\eta}(g^{\mu\lambda}g^{\nu\sigma}-g^{\mu\sigma}g^{\nu\lambda}) + \text{two more permutations}\Big]$$

This vertex is order g_s^2 (not g_s), so it contributes at the same perturbative order as two three-gluon vertices connected by a propagator. You will not need to evaluate this explicitly in Chapter 8 — Griffiths only uses it conceptually.

4 Putting It All Together: A Worked Comparison

Side-by-side: $e^-\mu^- \to e^-\mu^-$ vs. $q\,q \to q\,q$

Consider the simplest scattering process in each theory.

QED ($e^-\mu^-$ scattering, one-photon exchange):

$$\mathcal{M}_{\text{QED}} = \frac{g_e^2}{(p_1 - p_3)^2} \left[\bar{u}(3)\gamma^\mu u(1)\right] \left[\bar{u}(4)\gamma_\mu u(2)\right]$$

QCD (qq scattering, one-gluon exchange):

$$\mathcal{M}_{\text{QCD}} = \frac{g_s^2}{4} \frac{1}{(p_1 - p_3)^2} \left[\bar{u}(3)\gamma^\mu u(1)\right] \left[\bar{u}(4)\gamma_\mu u(2)\right] \times \underbrace{\sum_{a=1}^{8} \lambda^a_{j_3 j_1} \lambda^a_{j_4 j_2}}_{\text{color factor}}$$

The **only** differences are:

1. $g_e^2 \to g_s^2/4$ (different coupling, plus the $1/2$ at each vertex)
2. A sum over gluon types produces the **color factor**
3. The Dirac structure — all the spinors and gamma matrices — is *identical*

Why the color factor matters

The sum $\sum_a \lambda^a_{j_3 j_1} \lambda^a_{j_4 j_2}$ is the object that will determine whether quarks attract or repel in a given color configuration. In Section 8.4, Griffiths evaluates this sum for quarks in the color singlet vs. color octet, and shows that the singlet is maximally attractive. That's the QCD explanation for confinement (at the perturbative level): color singlets are the energetically preferred configurations.

5 Summary: What to Take Away

Key Points from Section 8.3

1. **QCD is QED with color.** The Dirac algebra is identical. The only new ingredients are color vectors, λ matrices, and gluon self-couplings.
2. **Color and spin factorize.** You can compute traces exactly as before, then multiply by the color factor at the end.
3. **The vertex factor is $-\frac{ig_s}{2}\lambda^a_{ji}\gamma^\mu$.** The λ^a_{ji} element tells you which color transitions are allowed and with what weight.
4. **The gluon propagator carries δ^{ab}.** Only one type of gluon flows through each internal line.
5. **Gluon self-interactions exist because $f^{abc} \neq 0$.** This is the fundamental difference

from QED. It leads to asymptotic freedom, confinement, and all the rich phenomenology of QCD.

6. **8 gluons, not 9**, because QCD is based on SU(3), not U(3). The color singlet combination is excluded.